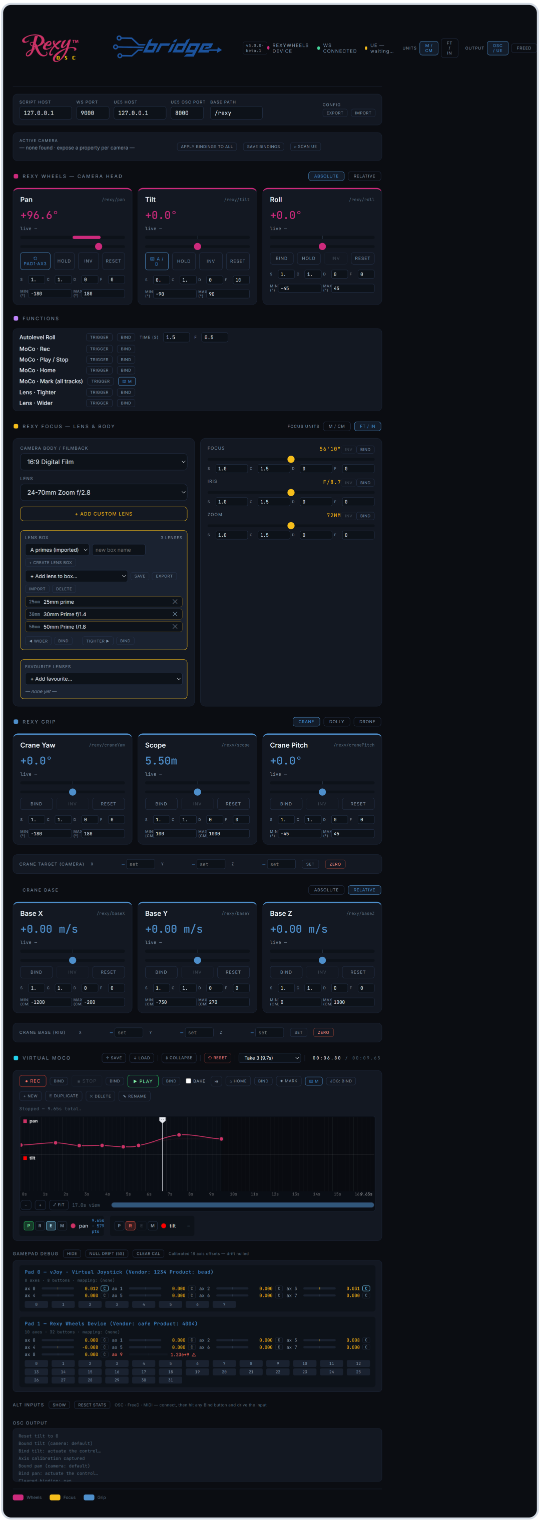
Rexy Bridge 3.0 — User Guide

A panel-by-panel tour of every control, in the order they appear in the app. Setup and UE connection are covered in **GETTING_STARTED.md**; this guide assumes you're connected and driving a camera.



Core concepts (read this first)

Binding. Almost every control has a **Bind** button. Click it, then actuate the input you want — a wheel, a stick, a keyboard key, a MIDI knob, or an OSC/FreeD channel. For continuous controls, push it the way you want to be "positive" first; for keys/buttons it captures a two-stage *increase* then *decrease*. Right-click a Bind button to clear it.

Per-camera bindings. Every camera you select keeps its **own** set of bindings. Switch cameras and your inputs re-map to that camera's saved layout.

S / C / D / F tuning. Wherever you see these four little boxes, they shape how an input drives the value:

- **S — Sensitivity / Speed.** Multiplies how fast the control moves the value. Works for analogue *and* keyboard/button inputs (higher = faster).
- **C — Curve.** Response gamma. `1.0` = linear; higher = "fluid-head" feel (gentle at first, faster as you push); lower = punchy.
- **D — Deadband.** Ignores tiny inputs near centre — kills stick/wheel drift. (Not relevant to keyboard/button inputs.)
- **F — Feather.** Smooths the start and stop so motion eases in and out instead of snapping.

Inv / Hold / Reset. Inv flips a binding's direction. **Hold** (on wheels) freezes that axis so its binding is ignored. **Reset** returns the control to centre/default.

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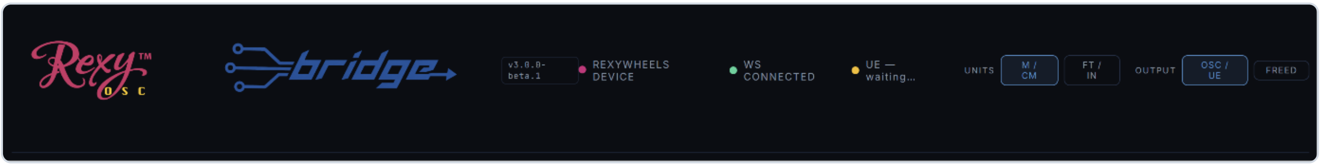
1. Top bar



- **Version badge** (`v3.0.0-beta.1`) — the running build. If it's missing or wrong after an update, press **Ctrl+Shift+R**.
- **HARDWARE** dot — your gamepad/wheels. Wakes when you click the window and press a physical button.
- **WS** dot — the internal app↔bridge link.
- **UE RC** dot — the Unreal Remote Control connection.
- **Units** — global metric (m/cm) vs imperial (ft/in) for distances and positions.

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2. Output mode & network

The **osc / ue** vs **freed** toggle chooses what Rexy Bridge drives, and the network fields beneath it change to match.

- **osc / ue** (default) — drives Unreal directly via Remote Control. The fields show the UE/OSC target host and port.

- **freed** — broadcasts a **FreeD** camera-tracking stream on the network so any FreeD-compatible app can consume it (no UE writes). The fields switch to the FreeD stream host/port.

Your connection settings, along with every binding and tuning value, save and reload from the Config controls — export a `.json` to move a whole setup between machines.

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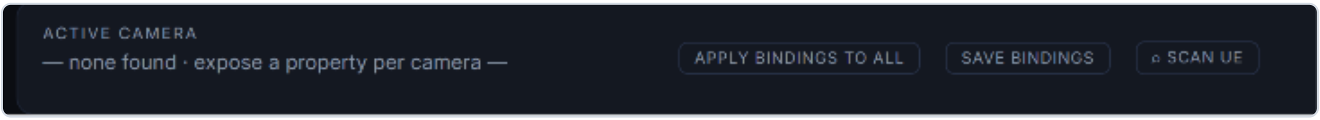
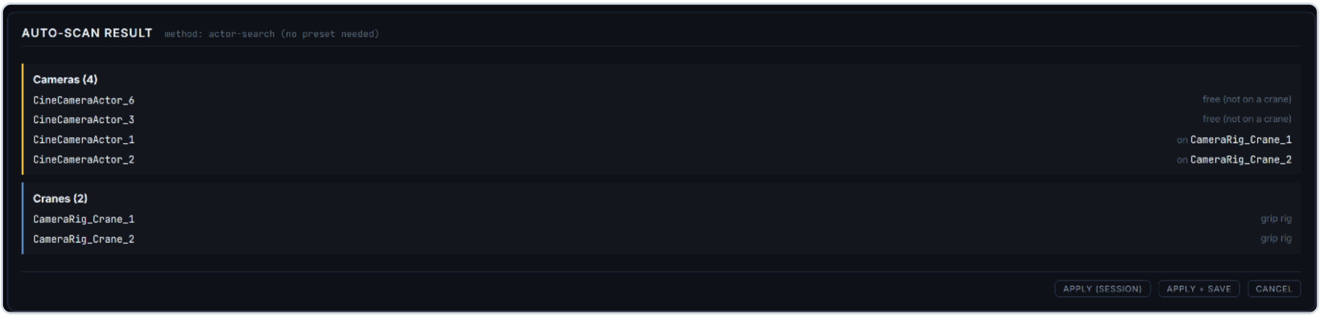
3. Cameras — Scan & select

Before scanning, the camera row is empty and waiting.

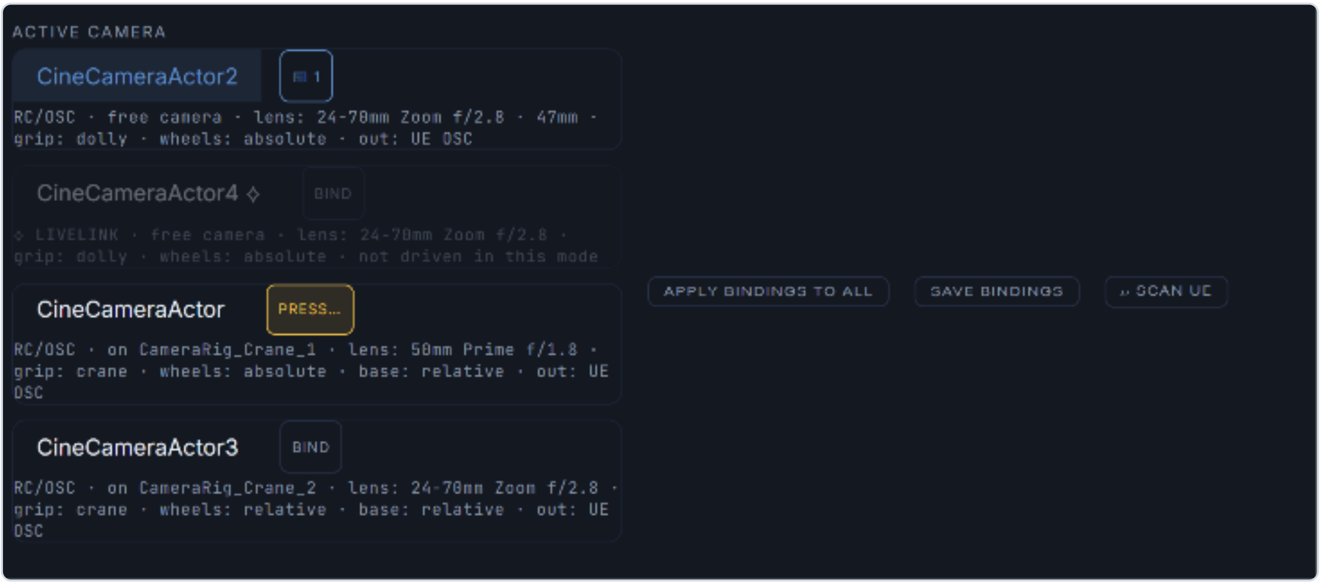


Click ⌕ **Scan UE** to auto-discover every Cine Camera Actor and CameraRig_Crane in your open level — no RexyControl preset, no `mappings.json` editing. Review the result, then **Apply** for this session or **Apply + Save** to persist it.



Discovered cameras appear as buttons. Click one to make it the **active** camera; its bindings and profile load.

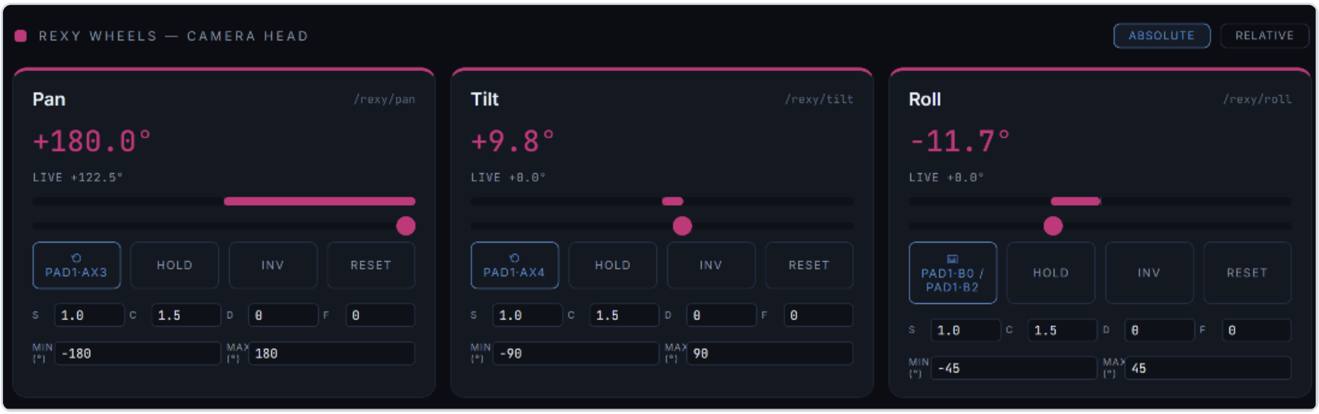
Each camera button can itself be **bound** to a key or gamepad button for hands-free switching mid-shot.



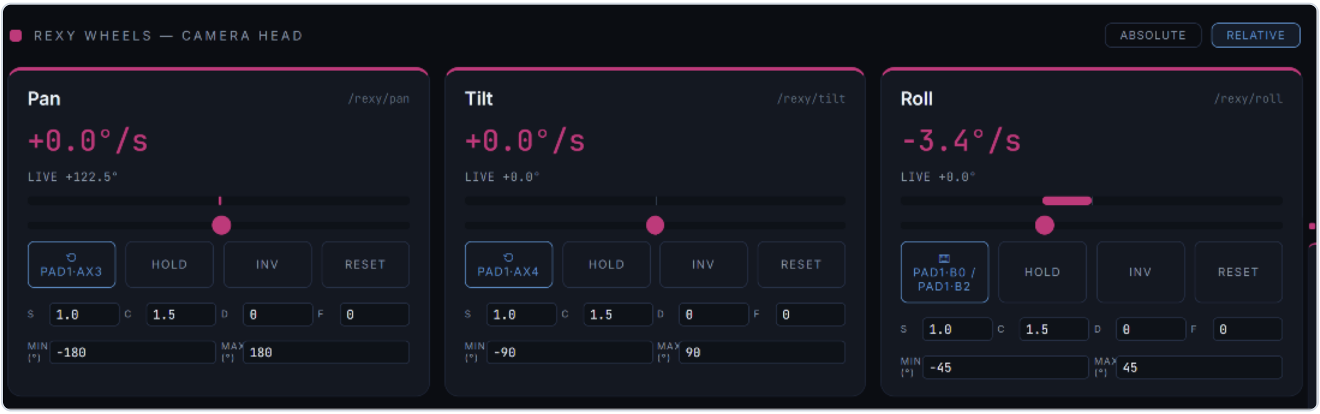
4. Rexy Wheels — camera head

The three head axes: **pan, tilt, roll**.

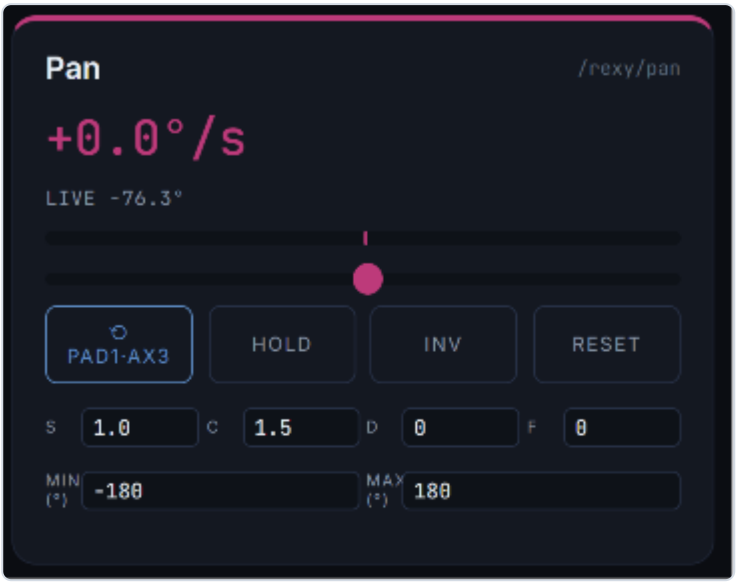
- **Absolute** — wheel position = camera angle.



- **Relative** — wheel deflection = rotation *speed*, centre = hold (endless-rotation, fluid-head style).

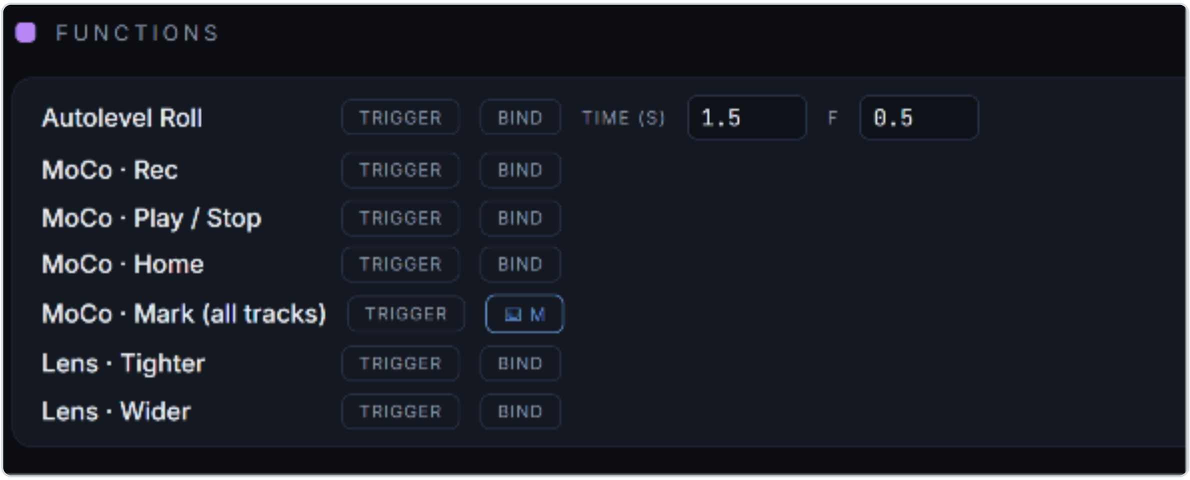


Each card has **Value** and **live —** readouts (commanded vs the camera's actual angle from UE), a **slider**, **Bind / Hold / Inv / Reset**, the **S / C / D / F** tuning row, and **Min / Max** to clamp the output range.



5. Functions

Bindable one-shot actions. Each card has a **Trigger** (fire it now) and a **Bind** (assign a key/button). Binding any copy of an action keeps every copy in sync.



- **Autolevel Roll** — smoothly returns roll to 0° over **Time** seconds, with an **F** easing.
- **MoCo · Rec / Play / Stop / Home / Mark** — transport actions, bindable so you never reach for the mouse mid-take. **Mark** drops a mark on **every bound track** at the playhead.
- **Lens · Tighter / Wider** — step to the next longer/shorter lens in the active **lens box**.

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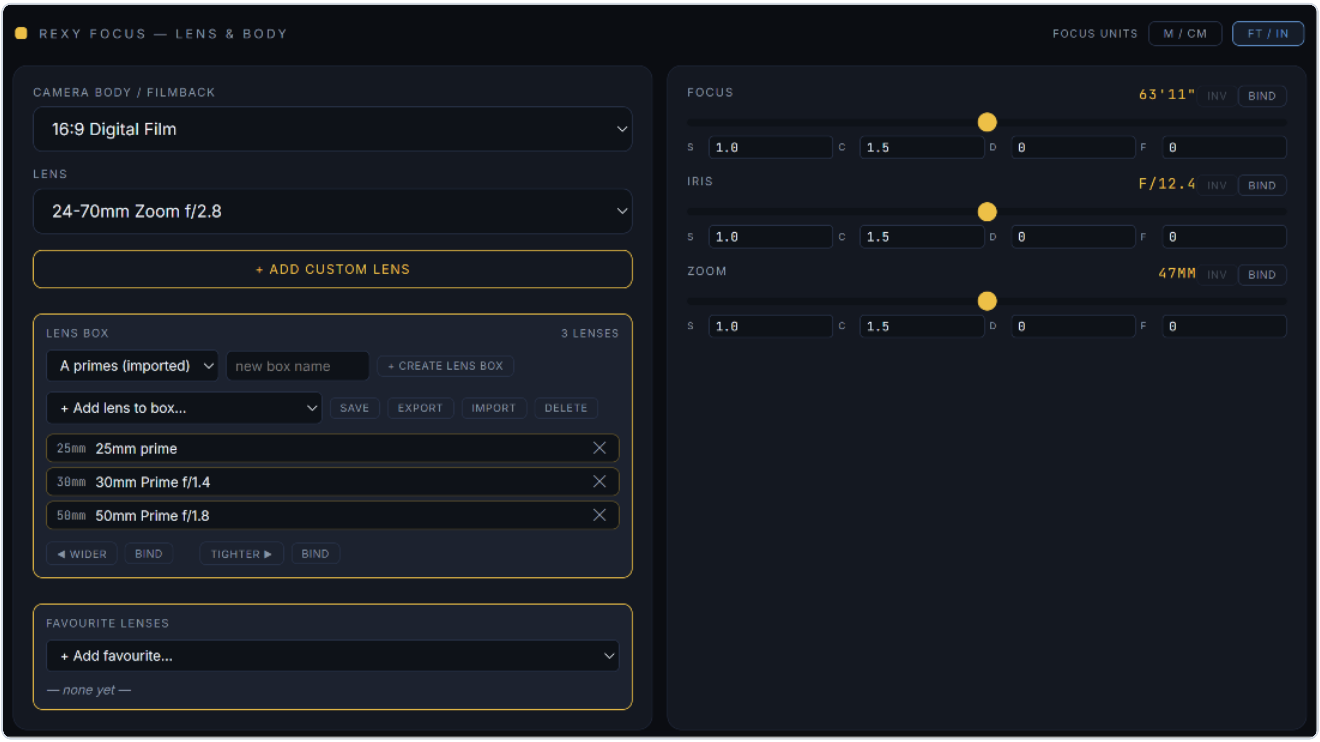
6. Rexy Focus — lens & body



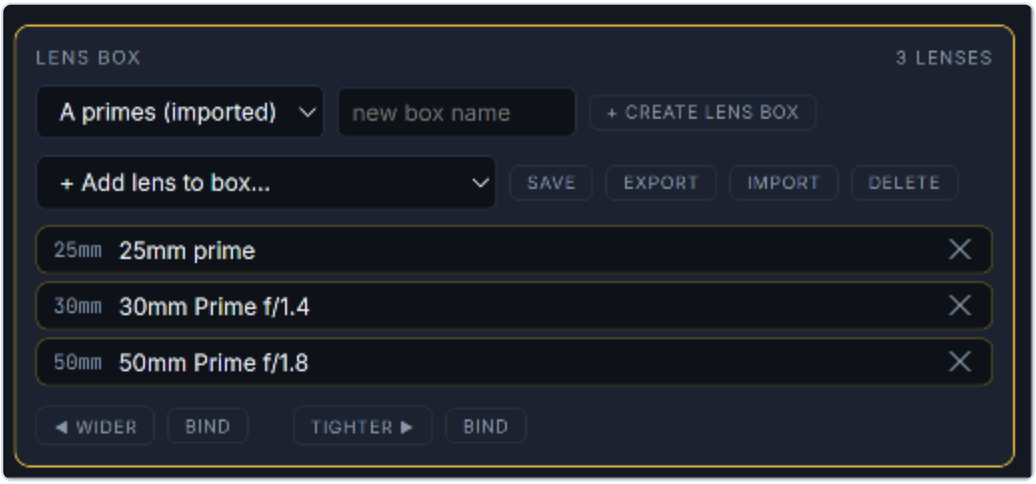
Lens & body

- **Camera body / filmback** — sensor/filmback preset.

- **Lens** — pick a lens; UE's LensSettings (focal, aperture, min-focus) update live.
- **+ Add custom lens** — define name, focal min/max, f-stop min/max, and **Min focus** (shown in cm or ft to match your Focus-units toggle). It's added to the list and applied.

Lens boxes

A **lens box** is a named kit — a list of lenses you cycle through on set.



- **Box selector** + **Create lens box** — make and switch between kits.
- **Add lens to box** dropdown — add any existing lens, or ✎ **+ Custom lens…** to define a new one that drops straight into the box.
- **Lens chips** — the box's lenses, always sorted by focal length; the active lens is highlighted. ✕ removes one.
- **◀ Wider / Tighter ▶** — step through the box by focal length (clamped at the ends). Each has a **Bind** for hardware control.
- **Save / Export / Import / Delete** — boxes export and import independently of the main config, so you can share a kit.

Favourite lenses

A global quick-jump list. Add any lens, then **Bind** a key/button to jump straight to it — independent of which box is loaded.

Focus / Iris / Zoom

Each has a slider, **Bind**, **Inv**, a live readout, and the full **S / C / D / F** tuning row — so a focus puller can bind a wheel or keys and tune the feel exactly like the head axes. Zoom locks automatically on a prime lens.

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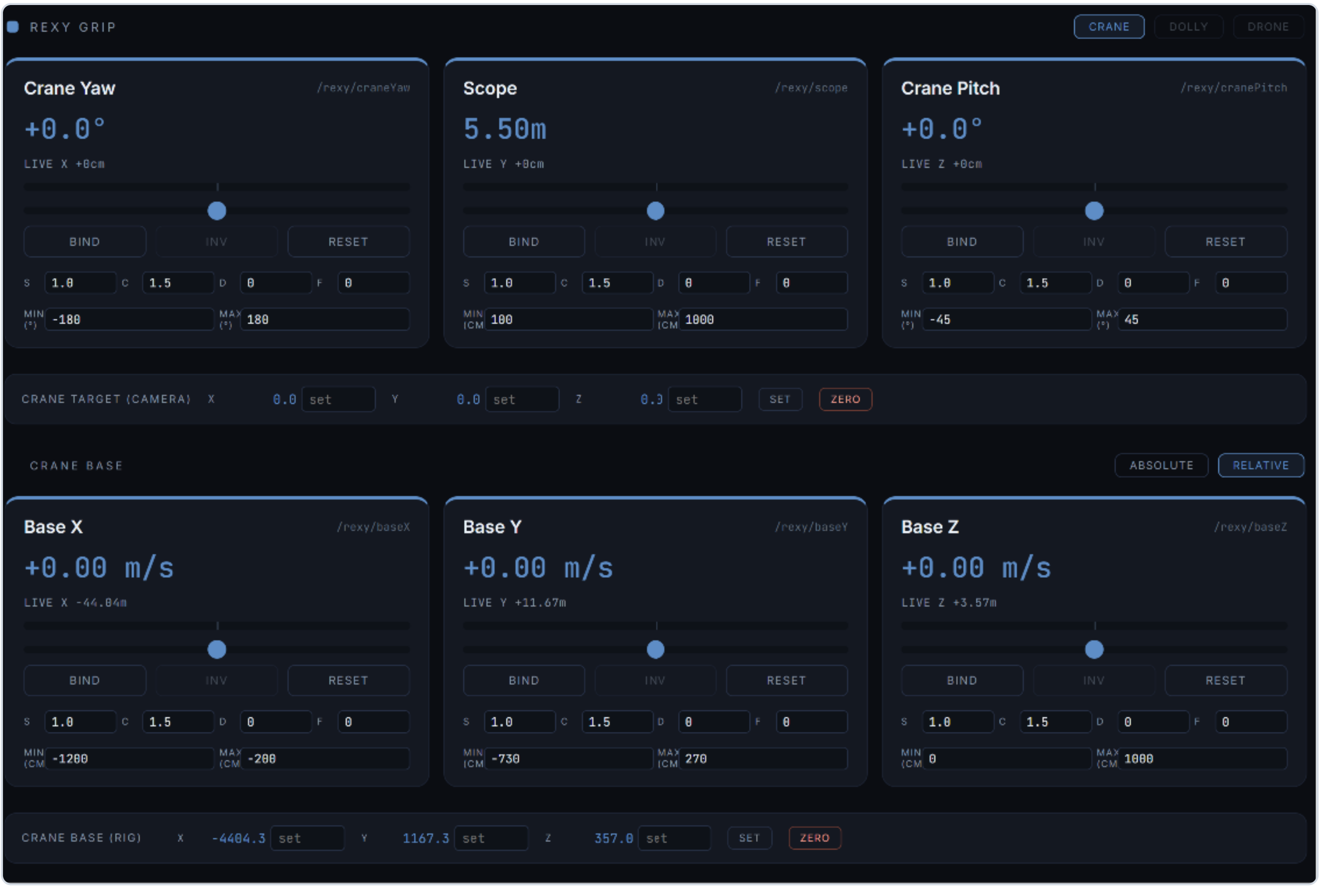
7. Grip — crane / dolly / drone



- **Crane** — the grip swings a jib arm around its base (yaw, pitch, scope/arm-length).
- **Dolly** — straight ground translation (X/Y/Z), no arm pivot.
- **Drone** — free flight through 3D space.

The grip cards (craneYaw, scope, cranePitch) use the same binding + S/C/D/F model as the wheels.

Crane Base

The base (baseX/Y/Z) drives the rig's pivot, with two modes:

- **Absolute** — slider position = base position between Min and Max.

- **Relative** — slider deviation = movement speed (m/s); centre = stop.

8. Position panels

Live **X / Y / Z** readouts, with **Set** (send a manual position; blank axes keep their current value) and **Zero** (reset to 0,0,0).

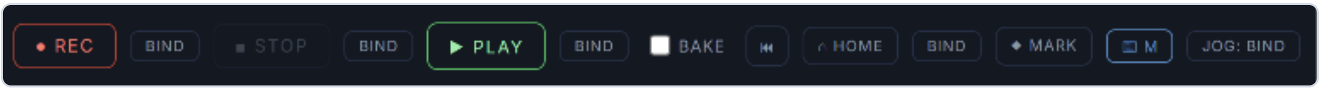
- **Crane target** — where the camera/target sits.

- **Crane base** — where the rig base sits.

9. Virtual MoCo

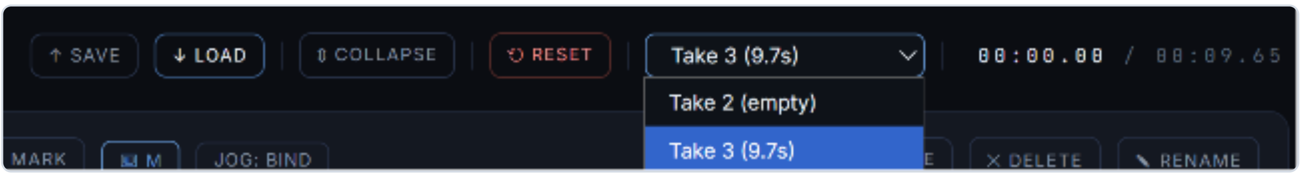
Record, edit, and play back moves across every bound parameter.

Transport



- ● **REC** / ■ **STOP** / ▶ **PLAY** — each with a **Bind** button. STOP leaves the playhead where it is; **Home** (or Rewind) returns it to the start.
- **Bake** — a 3-2-1 countdown on REC/PLAY so you can sync with UE's Take Recorder.
- ⌂ **HOME** — snaps play-armed tracks to their start values and the playhead to 0.
- ◆ **MARK** — drops a mark on every bound track at the playhead.
- **Jog: Bind** — bind an analogue wheel/axis (or keys) to **shuttle** the playhead: push from centre to scrub forward/back, speed proportional to deflection, centre = hold.

Takes



Multi-take management. Save/Load moves as `.rxmove` files. **Expand** stacks each track in its own lane; **Reset** clears everything.

Timeline & tracks

Colour-coded per track. Scrub by dragging the playhead, zoom with the scroll wheel, and drag anchor dots to reshape curves. Each track (left of the timeline) has four arm toggles:

- **P — Play** — plays back its recorded/marked data.
- **R — Record** — captures live input on the next REC (recording over a track auto-erases its marks).
- **E — Edit** — drag anchor dots to reshape. Available once a track has recorded data **or 2+ marks** — entering E on a marked-but-unrecorded track lays down draggable anchors that smooth the marks into **S-curves**.
- **M — Mark** — drops a fixed point at the playhead; the curve always passes through marks. Placing a mark auto-adds a start mark at t=0 so the move has a defined beginning.

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10. Gamepad Debug & calibration

- **Show / Hide** — toggles the live view of every axis and button on connected controllers.
- **Null Drift (5s)** — samples all axes for 5 seconds and stores the average as zero, nulling resting drift. **Clear Cal** removes it.

The per-axis **"C"** button opens the calibration wizard. Steps 1–3 capture the axis **max**, **min**, and **rest** so uneven hardware maps to a clean −1…+1.

The final **Tune response** step shapes the feel with a preset (Linear / Relaxed / Aggressive) or a custom curve, with a live waveform and curve preview. **Skip** jumps straight to the tuner; **Redo step** re-captures the current step.

11. Other inputs — OSC, MIDI, FreeD

- **OSC in** — accept OSC from external devices (e.g. the Rexy Wheels). Set **port**, an optional **prefix** filter, and the input **min/max** range. **Sample 10s** records every OSC message for ten seconds and exports a JSON log — the ground-truth capture for diagnosing a device.
- **MIDI** — **Enable MIDI** reads MIDI controls (knobs, faders, notes, pitch-bend) into the same input path as gamepads, s